JNTUK R23 REGULATION | SEMESTER I

Basic Civil & Mechanical Engineering

Study Module – Exam Ready Reference

UNIT II

SURVEYING

Subject Code: **20CE1101**

Regulation: **JNTUK R23**

Branch: **All Branches (B.Tech I Year)**

Prepared for: **End Semester Examination**

Topics: **Surveying Basics | Measurements | Bearings | Levelling | Contours**

Section 1: Introduction to Surveying

1.1 Definition of Surveying

Surveying is the art and science of making measurements of the natural and man-made features of the earth's surface and representing them on paper (maps or plans). It involves determining the relative positions of points on, above, or below the earth's surface by measuring distances, elevations, directions, and angles.

Surveying is one of the oldest and most fundamental branches of civil engineering. From ancient Egypt (where surveys were conducted to re-establish land boundaries after the Nile floods) to modern urban infrastructure planning, surveying has remained indispensable to human development.

> **Definition (IS 5511):** Surveying is the science of determining the position, in three-dimensional space, of natural and man-made features on or beneath the surface of the earth, these features being represented in analogue form as a map, chart, or plan or in digital form as a three-dimensional mathematical model.

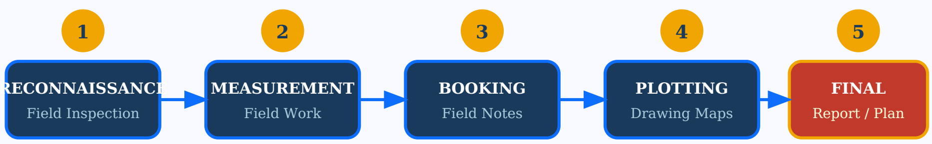
1.2 Basic Principles of Surveying

- **Working from whole to part:** Establish control points over the entire area first, then fill in detail surveys.
- **Locating a point by at least two measurements:** No point should be fixed by a single measurement; always use two independent measurements to detect errors.
- **Independent checking:** Every measurement must be independently checked to eliminate errors and blunders.
- **Accuracy vs Economy:** The degree of accuracy must be consistent with the nature and purpose of the survey.

1.3 Classification of Surveying

Basis of Classification	Types	Description
Nature of field	Land / Hydrographic / Astronomical	Based on area surveyed
Object of Survey	Topographic / Cadastral / City	Based on purpose
Instruments Used	Chain / Compass / Plane Table / Theodolite	Based on tools
Method of Survey	Triangulation / Traversing	Based on technique
Earth Curvature	Plane / Geodetic	Based on scale

FIG 1.1: SURVEYING PROCESS FLOW DIAGRAM



Section 2: Objectives of Surveying

2.1 Primary Objectives

The main objectives of surveying are as follows:

1. **To prepare maps and plans:** Surveying provides data to prepare topographic maps, site plans, cadastral maps, and engineering drawings needed for planning and design.
2. **To determine relative positions:** Establish precise horizontal and vertical positions of points for setting out engineering works like roads, railways, and buildings.
3. **To determine areas and volumes:** Calculate areas of land parcels and volumes of earthwork for projects like embankments, cuttings, and reservoirs.
4. **To establish boundaries:** Define land ownership and legal boundaries of properties and public lands.
5. **To set out engineering works:** Translate designs from paper to the ground accurately, such as road alignments, building positions, pipeline routes.
6. **To collect topographic data:** Gather information about the natural terrain, including hills, valleys, rivers, vegetation, and man-made features.
7. **To determine elevations:** Find heights of points to design drainage systems, road grades, and flood control measures.
8. **To monitor structural changes:** Detect deformations and movements of structures like dams, bridges, and tall buildings over time.

2.2 Applications of Surveying

Field	Application
Civil Engineering	Roads, bridges, dams, buildings, pipelines
Mining	Underground mine mapping, surface surveys
Agriculture	Land measurement, irrigation planning
Military	Topographic maps for defence strategy
Urban Planning	City maps, infrastructure layout
Archaeology	Site recording and excavation mapping
Remote Sensing	Satellite imagery interpretation
Navigation	Hydrographic charts, GPS systems

2.3 Importance of Surveying in Civil Engineering

No engineering project can be executed without prior surveying. Before any construction, engineers must know:

- The exact dimensions and boundaries of the site
- The topography (natural ground levels and slopes)
- The existing infrastructure (roads, utilities)
- The volume of earthwork required

Surveying bridges the gap between engineering design and actual construction on the ground.

Section 3: Horizontal Measurements

3.1 Definition

Horizontal measurement in surveying refers to the determination of horizontal distances between two points projected on a horizontal plane. The actual slope distance between two points on the ground is converted to its equivalent horizontal distance for accurate mapping and plan preparation.

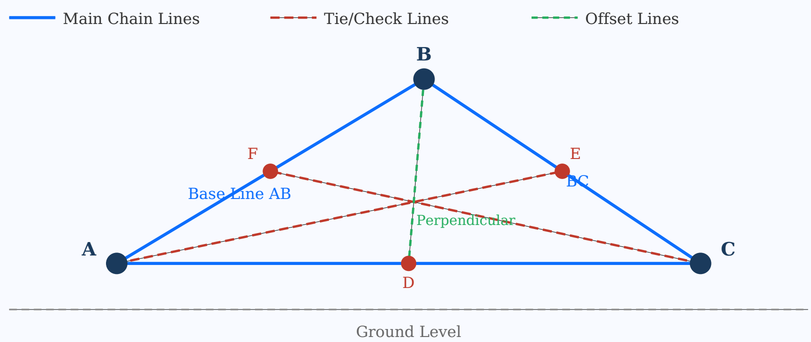
3.2 Chain Surveying

Chain surveying is the simplest and oldest method of surveying where the area is divided into a network of triangles and linear measurements alone are made with a chain or tape. It is suitable for open and flat areas with no obstacles.

Instruments Used:

- **Gunter's Chain (22 yards, 100 links):** Traditionally used for land surveys
- **Engineer's Chain (100 ft, 100 links):** Used in engineering surveys
- **Metric Chain (20 m or 30 m, each link = 20 cm):** Modern standard
- **Steel Tape:** For precise measurements
- **Arrows (Ranging Rods):** For marking chain lengths
- **Cross Staff / Optical Square:** For setting out right angles

FIG 3.1: CHAIN SURVEYING – NETWORK OF TRIANGLES



3.3 Tape Corrections

Key Formulas – Horizontal Measurement Corrections

1. Correction for Slope: $C_h = L(1 - \cos\theta)$ [where θ = angle of inclination]
2. Horizontal Distance from Slope: $H = L \times \cos\theta$
3. Correction for Temperature: $C_t = \alpha(T - T_0) \times L$
4. Correction for Pull/Tension: $C_p = (P - P_0)L / AE$
5. Sag Correction: $C_s = W^2L^3 / 24P^2$ [W = weight per unit length]
6. True Length = Measured Length + Corrections Applied

3.4 Advantages and Disadvantages of Chain Surveying

Advantages:

- Simple, inexpensive instruments required
- No angular measurement needed
- Easy to check for errors using check lines

- Suitable for small-area surveys

Disadvantages:

- Not suitable for hilly or wooded terrain
- Accumulation of errors over long distances
- Slow and labour intensive
- Not suitable for large areas

Section 4: Angular Measurements

4.1 Definition

Angular measurement in surveying is the determination of the angle between two survey lines (directions) in a horizontal plane. Angles are measured using a magnetic compass, theodolite, or total station. Angular measurements combined with linear measurements allow accurate location of points.

4.2 Types of Angles Measured

- **Horizontal Angle:** Angle between two lines projected onto a horizontal plane
- **Vertical Angle:** Angle in the vertical plane, above (elevation) or below (depression) horizontal
- **Deflection Angle:** Angle between a line and the extension of the preceding line
- **Interior Angle:** Angle inside a closed traverse polygon

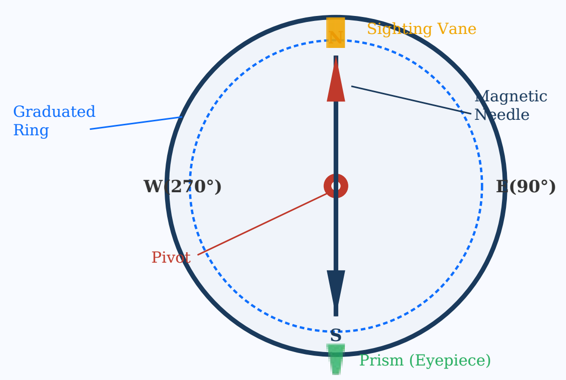
4.3 Prismatic Compass

The prismatic compass is the most commonly used instrument for angular measurements in the field. It measures magnetic bearings of survey lines directly.

Construction:

- **Circular Box:** Brass or aluminum casing, 85–110 mm diameter
- **Magnetic Needle:** Freely pivoted on a pivot; points to magnetic north
- **Graduated Ring:** Attached to needle, graduated 0°–360° clockwise from South (inverted reading)
- **Prism:** Eyepiece prism allows observer to read graduated ring while sighting the object
- **Sighting Vane:** Object vane with a vertical hair/wire for sighting
- **Brake Pin / Agate Cap:** Lifts needle off pivot to prevent wear when not in use

FIG 4.1: PRISMATIC COMPASS – CROSS-SECTION VIEW



Working Principle:

1. Set up the compass on the tripod over the survey station
2. Level the instrument using the ball and socket joint
3. Release the brake pin to allow the needle to swing freely
4. Sight the ranging rod at the far station through the sighting vane
5. Read the bearing through the prism, which shows the value of the graduated ring at the point aligned with the sighting vane
6. Record the reading; this gives the whole circle bearing (WCB) of the line

Check Sum – Closed Traverse

Sum of all interior angles of a closed traverse = (2n − 4) × 90°

where n = number of sides in the traverse

Section 5: Bearings – Introduction

5.1 Definition of Bearing

The bearing of a line is defined as the horizontal angle made by the line with a reference direction (north). Bearings are always measured in a clockwise direction from the reference meridian and are used to define the direction of survey lines.

5.2 Reference Meridians

- **True Meridian:** The line joining the geographic north and south poles passing through the observer. True bearing is the most accurate.
- **Magnetic Meridian:** Direction pointed by a freely suspended magnetic needle (towards magnetic north).
- **Arbitrary Meridian:** Any convenient direction assumed as reference for local surveys.
- **Grid Meridian:** The central meridian of a grid zone used in coordinate systems.

5.3 Magnetic Declination

Magnetic Declination (δ): The horizontal angle between the true meridian and the magnetic meridian at any place. It varies with location and time.

Eastern Declination: Magnetic north is east of true north → True Bearing = Magnetic Bearing – δ

Western Declination: Magnetic north is west of true north → True Bearing = Magnetic Bearing + δ

Section 6: Types of Bearings

6.1 Whole Circle Bearing (WCB)

In the Whole Circle Bearing system, the bearing is measured from the north direction in the clockwise direction, continuously from 0° to 360°. This is the system used by the prismatic compass.

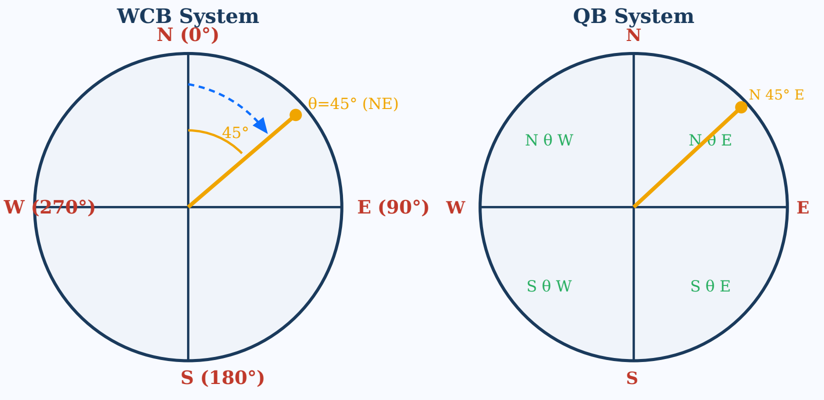
- North = 0° (or 360°)
- East = 90°
- South = 180°
- West = 270°

6.2 Quadrantal Bearing (QB) / Reduced Bearing (RB)

In the Quadrantal Bearing system, the bearing is measured from north or south, whichever is nearer, towards east or west. Values range from 0° to 90°. The surveyor's compass uses this system.

The bearing is written as: **N θ E, N θ W, S θ E, or S θ W**

FIG 6.1: WCB VS QB BEARING SYSTEM COMPARISON



6.3 Conversion: WCB ↔ QB

Quadrant	WCB Range	QB Formula	Example
I (NE)	0° – 90°	QB = WCB (N _ E)	WCB 60° → N 60° E
II (SE)	90° – 180°	QB = 180° – WCB (S _ E)	WCB 130° → S 50° E
III (SW)	180° – 270°	QB = WCB – 180° (S _ W)	WCB 220° → S 40° W
IV (NW)	270° – 360°	QB = 360° – WCB (N _ W)	WCB 310° → N 50° W

6.4 Fore Bearing and Back Bearing

Fore Bearing and Back Bearing

```
Back Bearing (WCB) = Fore Bearing ± 180°
If FB < 180°: BB = FB + 180°
If FB > 180°: BB = FB - 180°
For QB: Change N to S (or S to N) and E to W (or W to E)
```

Section 7: Levelling

7.1 Definition of Levelling

Levelling is the operation of determining the difference in elevation (height) between two or more points on the earth's surface. It is a type of surveying in which measurements are made in the vertical plane.

> **Reduced Level (RL):** The height of a point above (or below) an assumed datum. In India, the datum is Mean Sea Level (MSL) at Karachi, published as the Great Trigonometrical Survey (GTS) bench mark.

7.2 Important Terminologies

Term	Abbreviation	Definition
Reduced Level	RL	Height of a point above the datum
Benchmark	BM	A fixed reference point of known elevation
Back Sight	BS	First staff reading taken after setting up the instrument
Fore Sight	FS	Last staff reading before shifting the instrument
Intermediate Sight	IS	Any staff reading between BS and FS
Height of Instrument	HI	RL of the line of sight (RL of BM + BS reading)
Change Point	CP	A point where both FS and BS readings are taken
Rise	–	Difference in elevation when new RL > previous RL
Fall	–	Difference in elevation when new RL < previous RL

Fundamental Levelling Equations

```
Height of Instrument (HI) = RL of BM + BS
RL of next point = HI − FS (or HI − IS)
Rise or Fall = Previous Staff Reading − Present Staff Reading
If result is (+) → Rise; If result is (−) → Fall
RL of new point = RL of previous point + Rise (or − Fall)
Check: ΣBS − ΣFS = ΣRise − ΣFall = Last RL − First RL
```

Section 8: Levelling Instruments

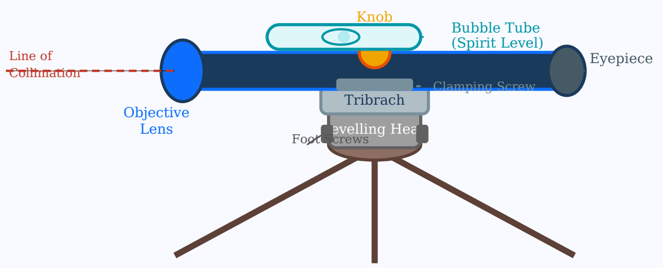
8.1 Dumpy Level

The dumpy level is the most commonly used levelling instrument. It has a telescope rigidly fixed to the level tube and cannot be rotated about its longitudinal axis. It is robust, simple, and gives accurate results.

Parts of the Dumpy Level:

- **Telescope:** Used to sight the levelling staff. Contains objective lens, eyepiece, cross-hairs (diaphragm), and focusing knob.
- **Bubble Tube (Spirit Level):** Cylindrical glass tube with liquid and air bubble. When the bubble is centred, the instrument is level.
- **Tribrach:** Three-arm base plate with levelling screws (foot screws) to level the instrument.
- **Levelling Head:** Connects the tribrach to the tripod; contains foot screws.
- **Tripod:** Three-legged support to hold the instrument stable on the ground.
- **Objective Lens:** Collects light and focuses the image of the staff.
- **Diaphragm (Cross-hairs):** Glass plate with etched horizontal and vertical lines for accurate sighting.

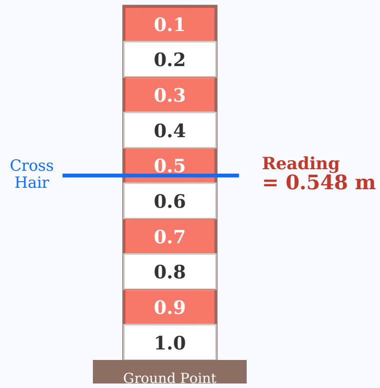
FIG 8.1: DUMPY LEVEL – LABELED DIAGRAM



8.2 Levelling Staff

The levelling staff is a graduated rod used to read the difference in elevation. It is held vertically at the point whose elevation is to be determined.

FIG 8.2: LEVELLING STAFF READING



8.3 Other Levelling Instruments

- **Tilting Level:** Telescope can be tilted slightly about a horizontal axis using a tilting screw; gives high accuracy.
- **Auto Level (Automatic Level):** Uses a compensator to automatically maintain a horizontal line of sight.
- **Digital Level:** Reads staff electronically using bar-coded staffs; eliminates human reading error.
- **Hand Level:** Simple, hand-held instrument for rough levelling checks.

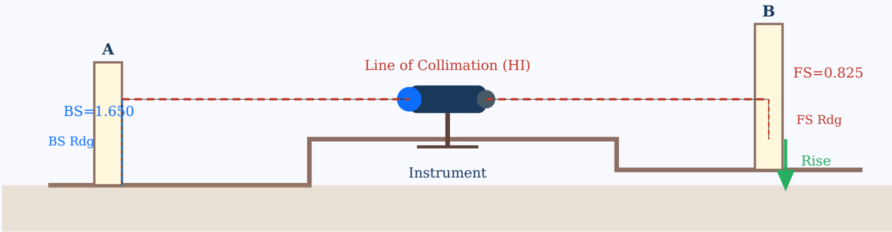
Section 9: Methods of Levelling

9.1 Types of Levelling

A. Simple Levelling

When the difference in elevation between two points can be determined from a single instrument setting, it is called simple levelling. The instrument is set up at a convenient position between the two points (A and B), and staff readings are taken at both.

FIG 9.1: SIMPLE LEVELLING SETUP



Simple Levelling Calculation

```
HI = RL of A + BS = RL_A + 1.650
RL of B = HI - FS = HI - 0.825
Difference in Elevation = BS - FS = 1.650 - 0.825 = 0.825 m (Rise)
```

B. Differential (Fly) Levelling

When the two points are far apart or not intervisible, the instrument is set up at multiple positions and the levels are carried forward using change points. This is called differential levelling or fly levelling.

C. Profile Levelling (Longitudinal Sectioning)

Levels are taken along a line (e.g., a road or railway alignment) at regular intervals to plot the longitudinal profile of the ground for design purposes.

D. Cross-Section Levelling

Staff readings are taken at right angles to the main alignment at each chainage to determine the cross-sectional shape of the ground.

E. Reciprocal Levelling

Used when two points are separated by an obstacle (river, valley). The instrument is set up on each bank alternately and readings taken; the average of the two results gives the true difference in elevation, eliminating errors due to curvature, refraction, and collimation.

9.2 Comparison of Levelling Methods

Method	Best Used For	Accuracy	No. of Instrument Positions
Simple Levelling	Short distances, visible points	High	Single
Differential Levelling	Long distances	High	Multiple
Profile Levelling	Road / Pipeline routes	Medium	Multiple
Cross-section Levelling	Earthwork volumes	Medium	Multiple
Reciprocal Levelling	Across rivers/valleys	Very High	Two (one each side)

Section 9 (Contd.): Methods of Computing RL

9.3 Height of Instrument (HI) Method

In the HI method (also called Collimation method), the RL of the instrument line of sight (HI) is computed for each instrument position and used to find the RL of all intermediate and fore sight points.

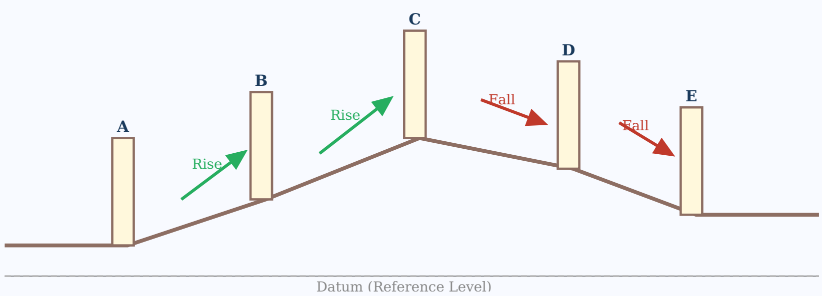
- **Step 1:** HI = RL of BM + BS (Back Sight reading)
- **Step 2:** RL of all IS points = HI – IS reading
- **Step 3:** RL of FS point = HI – FS reading
- **Step 4:** Repeat for each instrument position
- **Check:** ΣBS – ΣFS = Last RL – First RL

9.4 Rise and Fall Method

In the Rise and Fall method, the difference between consecutive staff readings determines whether there is a rise or fall from one point to the next. This gives a complete check on every reading.

- **Rise:** Previous Reading – Present Reading = (+) → Point is higher
- **Fall:** Previous Reading – Present Reading = (–) → Point is lower
- **RL of new point = RL of previous point + Rise (or – Fall)**
- **Arithmetic Check:** ΣBS – ΣFS = ΣRise – ΣFall = Last RL – First RL

FIG 9.2: RISE AND FALL METHOD – VISUAL REPRESENTATION



9.5 Comparison: HI Method vs Rise & Fall Method

Feature	HI Method	Rise & Fall Method
Speed	Faster	Slower
Error Check	Checks only BS and FS	Full arithmetic check on every IS
Suitability	Profile levelling with many IS	Precise levelling, few points
Simplicity	Simpler calculations	More systematic
Detection of errors in IS	Not directly possible	Easily detected

Section 10: Simple Problems on Levelling

Problem 1: Height of Instrument (HI) Method

Given Data: The following consecutive readings were taken with a level. The instrument was shifted after the 4th and 7th readings. RL of A (BM) = 100.000 m.
Readings: 2.228, 1.606, 0.988, 1.904, 1.264, 2.146, 1.428, 0.820, 1.682

Step-by-Step Solution using HI Method:

Point	BS	IS	FS	HI	RL	Remarks
A (BM)	2.228	–	–	102.228	100.000	BM
1	–	1.606	–	102.228	100.622	IS
2	–	0.988	–	102.228	101.240	IS
3 (CP)	1.904	–	1.904	102.228	100.324	CP
–	–	–	–	102.228	–	New HI: 100.324 + 1.904 = 102.228
4	–	1.264	–	102.228	100.964	IS
5	–	2.146	–	102.228	100.082	IS
6 (CP)	1.428	–	1.428	102.228	100.800	CP
7	–	0.820	–	102.228	101.408	IS
B	–	–	1.682	102.228	100.546	FS
Sum	**5.560**	–	**5.014**	–	–	–

Arithmetic Check (HI Method)

$\Sigma BS - \Sigma FS = 5.560 - 5.014 = +0.546$ m

Last RL − First RL = 100.546 − 100.000 = +0.546 m ✓ (Check Satisfied)

Problem 2: Rise and Fall Method

Given Data: Staff readings on a line of levels in the order taken: 0.855, 1.545, 1.025, 2.250, 1.100, 3.500, 0.600. RL of first point = 100.000 m.

Point	BS	IS	FS	Rise (+)	Fall (-)	RL
A	0.855	–	–	–	–	100.000
B	–	1.545	–	–	0.690	99.310
C	–	1.025	–	0.520	–	99.830
D (CP)	1.100	–	2.250	–	1.225	98.605
E	–	–	–	–	–	–
F (CP)	0.600	–	3.500	–	2.400	96.205

G	–	–	0.600	–	–	96.205
Sum	**2.555**	–	**6.350**	**0.520**	**4.315**	–

Arithmetic Check (Rise & Fall Method)

$\Sigma BS - \Sigma FS = 2.555 - 6.350 = -3.795$ m

$\Sigma Rise - \Sigma Fall = 0.520 - 4.315 = -3.795$ m

Last RL – First RL = 96.205 – 100.000 = –3.795 m ✓ (Check Satisfied)

Section 11: Simple Problems on Bearings

Problem 1: Conversion of WCB to QB

Convert the following WCB values to Quadrantal Bearings:

Line	WCB	Quadrant	QB (Reduced Bearing)
AB	45°30'	NE (I)	N 45°30' E
BC	125°15'	SE (II)	S (180° – 125°15') E = S 54°45' E
CD	210°40'	SW (III)	S (210°40' – 180°) W = S 30°40' W
DE	315°20'	NW (IV)	N (360° – 315°20') W = N 44°40' W

Problem 2: Calculate Back Bearings

Find the back bearings for the following fore bearings:

Line	Fore Bearing (WCB)	Back Bearing (WCB)	Method
PQ	48°30'	228°30'	FB < 180°, so BB = FB + 180°
QR	252°15'	72°15'	FB > 180°, so BB = FB – 180°
RS	180°00'	0°00'	FB = 180°, BB = 0°
ST	90°00'	270°00'	FB = 90°, BB = 90° + 180° = 270°

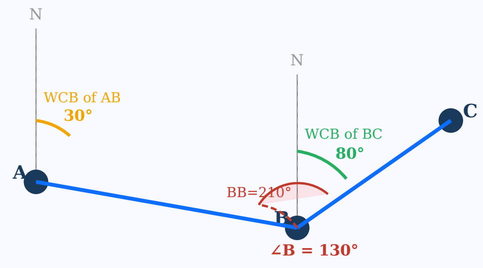
Problem 3: Included Angles from Bearings

Problem: The bearings of lines AB, BC, CD, and DE of a closed traverse are 30°, 80°, 130°, and 270° (WCB). Find the included angles.

The included angle at a station = Back Bearing of the previous line – Fore Bearing of the next line (add 360° if negative).

Station	Back Bearing of Incoming Line	Fore Bearing of Outgoing Line	Included Angle
B	BB of AB = 30° + 180° = 210°	FB of BC = 80°	210° – 80° = 130°
C	BB of BC = 80° + 180° = 260°	FB of CD = 130°	260° – 130° = 130°
D	BB of CD = 130° + 180° = 310°	FB of DE = 270°	310° – 270° = 40°

FIG 11.1: BEARING CALCULATION DIAGRAM


N
N
WCB of AB
30°
WCB of BC
80°
C
A
BB=210°
B
∠B = 130°

Section 12: Contour Mapping

12.1 Introduction to Contours

A contour is an imaginary line on the ground joining points of equal elevation (RL). On a map, contour lines are curved lines that represent the three-dimensional shape of the terrain in two dimensions.

The vertical distance between two consecutive contour lines is called the **Contour Interval (CI)**. The horizontal distance between contour lines on a map is called the **Horizontal Equivalent (HE)**.

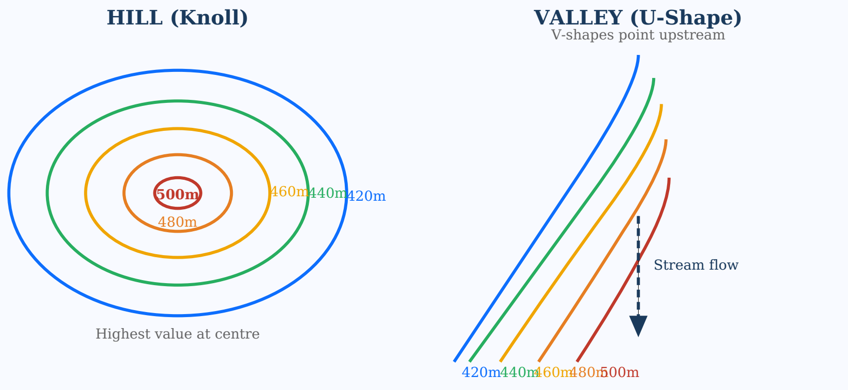
Contour Formulas

```
Contour Interval (CI) = Vertical distance between two consecutive contours
Horizontal Equivalent (HE) = (CI / Slope gradient) × Map scale factor
Slope = CI / HE × 100 (in %)
Grade = CI / HE (as a ratio 1:n)
```

12.2 Characteristics of Contour Lines

- Contour lines are always closed curves (they close on themselves, though not necessarily on the map sheet).
- Two contour lines of different elevations never cross each other (except for overhanging cliffs).
- Closer contour lines indicate steep slopes; widely spaced contours indicate gentle slopes.
- Contour lines cross ridges and valleys at right angles.
- In a valley, contours form V-shapes pointing upstream (uphill).
- On a ridge (hill), contours form U-shapes pointing downhill.
- A hill is represented by a series of closed contours, with higher values at the centre.
- A depression (hollow) is represented like a hill but with inward-facing tick marks.
- Uniform slope = equally spaced contours; concave slope = closer at top; convex slope = closer at bottom.

FIG 12.1: CONTOUR LINES – HILL AND VALLEY REPRESENTATION



12.3 Methods of Contouring

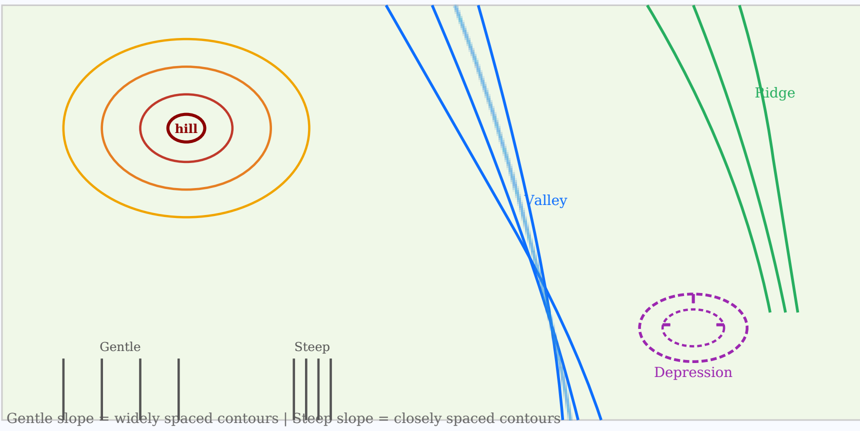
- **Direct Method:** Points of equal elevation are located directly in the field and joined on the map. More accurate but slow and expensive.
- **Indirect Method (Grid Method):** A grid is set up over the area, staff readings are taken at each grid intersection, and contours are interpolated between the grid points. Faster and commonly used.
- **Cross-section Method:** Sections are taken at regular intervals and contours interpolated from cross-section data. Useful for route surveys.

Section 12 (Contd.): Contour Maps and Applications

12.4 Applications of Contour Maps

- **Determination of intervisibility:** Finding whether two points are visible to each other across terrain.
- **Tracing a contour gradient:** Locating a route of constant slope for roads, canals, and pipelines.
- **Drawing cross-sections:** Estimating earthwork volumes for cuttings and embankments.
- **Reservoir capacity:** Calculating the volume of water stored in a reservoir at different levels.
- **Selection of dam sites:** Finding the best location for maximum catchment with minimum dam wall length.
- **Watershed delineation:** Identifying drainage basins and catchment areas.
- **Slope analysis:** Determining average and maximum slopes for erosion studies.
- **Military planning:** Identifying terrain features for strategic planning.

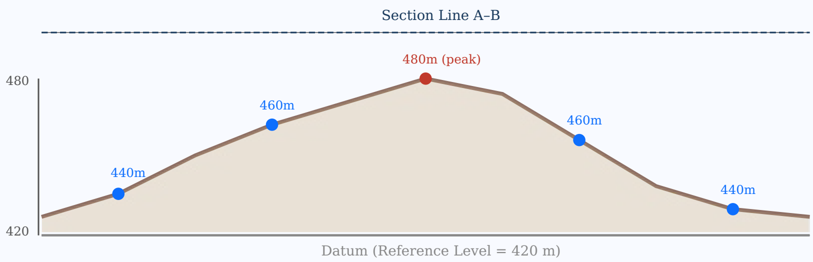
FIG 12.2: CONTOUR MAP SHOWING VARIOUS TERRAIN FEATURES



12.5 Cross-Section from Contours

A cross-section is a vertical section cut through the terrain along a specified line, derived from contour data. Engineers use cross-sections to calculate earthwork volumes (cut and fill) required for road and railway construction.

FIG 12.3: CROSS-SECTION DERIVED FROM CONTOUR MAP



Section 13: Questions & Answers

2-Mark Questions & Answers

2M Q1. Define Surveying and state its objectives.

Surveying is the science and art of determining the relative positions of points on, above, or beneath the earth's surface by measuring distances, angles, and elevations, and representing them on maps or plans. The primary objectives are: (1) to prepare maps and plans for engineering projects, (2) to determine areas, volumes, and differences in elevation, (3) to set out engineering works on the ground, and (4) to establish boundaries of land.

2M Q2. What is a bearing? Differentiate between WCB and QB.

A **bearing** is the horizontal angle made by a survey line with a reference meridian (north). In the **Whole Circle Bearing (WCB)** system, angles are measured clockwise from north through 0° to 360°. In the **Quadrantal Bearing (QB)** system, angles are measured from north or south (whichever is nearer) towards east or west, giving values between 0° and 90°. Example: WCB 135° = QB S 45° E.

2M Q3. Define Back Bearing and give the formula.

Back Bearing (BB) is the bearing of a line in the reverse direction. If the fore bearing of line AB is known, the back bearing gives the direction from B to A. Formula: If FB < 180°, BB = FB + 180°. If FB > 180°, BB = FB – 180°. In QB, simply interchange N with S and E with W to get the back bearing.

2M Q4. What is a Benchmark (BM)? Name its types.

A **Benchmark (BM)** is a fixed reference point of known elevation used as a starting point for levelling surveys. Types: (1) **GTS (Great Trigonometrical Survey) Benchmark** – established by the Survey of India, permanent and highly accurate; (2) **Permanent Benchmark** – established by public works departments; (3) **Temporary Benchmark** – established for a specific survey project; (4) **Arbitrary Benchmark** – assumed datum used for local surveys.

2M Q5. Define contour interval and horizontal equivalent.

The **Contour Interval (CI)** is the vertical distance between two consecutive contour lines on a map. It remains constant for a given map. The **Horizontal Equivalent (HE)** is the horizontal distance between two consecutive contour lines as measured on the map. For steep slopes, HE is small (closely spaced contours); for gentle slopes, HE is large (widely spaced contours). Slope = CI / HE.

Section 13 (Contd.): 5-Mark & 10-Mark Questions

5M Q6. Explain levelling with a neat diagram. State the purpose and instruments used.

Definition: Levelling is the operation of determining the difference in elevation between two or more points on the earth's surface. It is performed to find the relative heights of points and to transfer elevations from one point to another.

Purpose of Levelling: (1) To determine the elevation of existing structures and natural features; (2) To set out engineering works like roads, pipelines, canals at desired elevations; (3) To calculate earthwork volumes; (4) To draw longitudinal and cross-sections for design; (5) To check the gradient of drainage channels and sewers.

Instruments Used:

- **Level:** An optical instrument (Dumpy Level, Auto Level, Digital Level) that establishes a horizontal line of sight.
- **Levelling Staff:** A graduated wooden or aluminium rod held vertically at the point. Readings are taken through the telescope cross-hair.
- **Tripod:** Three-legged support for the level instrument.

Working Procedure (Simple Levelling):

1. Set up the instrument midway between points A and B; level carefully using foot screws.
2. Hold the staff vertically at A (BM); read the back sight (BS = 1.545 m); calculate $HI = RL_A + BS$.
3. Transfer staff to B; read the fore sight (FS = 0.720 m); calculate $RL_B = HI – FS$.
4. Difference in elevation = BS – FS = 1.545 – 0.720 = +0.825 m (Rise from A to B).

Arithmetic Check: $\Sigma BS – \Sigma FS$ = Last RL – First RL. This confirms the accuracy of all readings taken.

Applications: Road design, drainage, building construction, canal alignment, and flood control planning all require accurate levelling data.

10M Q7. Explain contour mapping with a neat diagram. What are the characteristics and uses of contour lines?

Definition: A contour map is a topographic map that shows the three-dimensional features of the earth's surface using contour lines on a two-dimensional plane.

Contour Line: An imaginary line on the ground joining all points of equal elevation. The vertical distance between consecutive contour lines is the Contour Interval (CI).

Methods of Contouring:

- **Direct Method:** Points of equal elevation are physically located in the field and plotted. Accurate but slow.
- **Indirect Method (Grid):** A grid pattern is surveyed, elevations recorded at intersections, and contours interpolated. Fast and commonly used.
- **Tachometric Method:** Uses a theodolite and staff readings to determine position and elevation simultaneously.

Characteristics of Contour Lines:

1. All points on a contour line have the same elevation.
2. Contour lines close on themselves (may close outside the map boundary).
3. Two contour lines of different elevations cannot intersect each other.
4. Closely spaced contours = steep terrain; widely spaced = gentle slope.
5. Contour lines cross ridges and valleys perpendicularly.
6. A hill is shown by closed contours with increasing values toward the centre.
7. A depression is shown by closed contours with inward-pointing tick marks.
8. A valley shows V-shaped contours pointing upstream (toward higher ground).
9. A ridge shows U-shaped contours opening toward lower elevation.

Uses of Contour Maps:

- Determination of intervisibility between two points (for communication towers, etc.).

- Tracing routes of constant gradient for roads, irrigation channels, and pipelines.
- Estimation of reservoir capacity (volume of water) at different water levels.
- Selection of dam sites to maximize storage with minimal structural length.
- Calculation of earthwork volumes (cut and fill) for roads and embankments.
- Delineation of watershed boundaries and drainage basins for hydrology studies.
- Military use: terrain analysis, field of fire, line of sight, route planning.

Section 13 (Contd.): Additional Questions

5M Q8. Explain the construction and working of a Dumpy Level with a neat labeled diagram.

Dumpy Level: A levelling instrument in which the telescope is permanently attached to the level tube and cannot be rotated about its vertical or horizontal axis relative to the levelling head.

Main Parts:

- **Telescope:** Main sighting element with objective lens, eyepiece, diaphragm (cross-hairs), and focusing screw. Magnification typically 20–40×.
- **Bubble Tube (Plate Level):** Glass tube filled with spirit; when the bubble is centred, the instrument axis is horizontal.
- **Tribrach:** Three-armed levelling head resting on three foot screws for precise levelling.
- **Levelling Screws (Foot Screws):** Three screws used to bring the bubble to centre and thereby level the instrument.
- **Tripod:** Three-legged support providing stability on uneven ground.
- **Clamp and Tangent Screw:** For locking and fine horizontal rotation of the telescope.

Working Principle:

1. Set up the tripod and mount the level; roughly level using the bull's-eye bubble.
2. Use three foot screws to centre the elongated bubble tube precisely.
3. Focus the telescope on the levelling staff at the target point.
4. Read the staff at the central horizontal cross-hair; record the reading.
5. The reading gives the height of the staff at that point below the line of collimation (HI).
6. RL of the point = HI – Staff Reading.

Advantages: Simple, robust, and gives consistent results when properly adjusted. **Disadvantages:** Requires re-levelling at each setup; less flexible than tilting or auto levels.

5M Q9. Describe the Rise and Fall method of levelling with an example. State its advantages.

Rise and Fall Method: In this method, the difference in elevation between consecutive points is determined by comparing their staff readings. No Height of Instrument is calculated.

Rules:

- If previous reading > present reading → Rise (+) → New RL = Previous RL + Rise
- If previous reading < present reading → Fall (–) → New RL = Previous RL – Fall

Arithmetic Check: ΣBS – ΣFS = ΣRise – ΣFall = Last RL – First RL. All three differences must be equal. This is a complete check on every reading.

Example: Given RL of A = 50.000 m, Staff readings: A(BS)=1.500, B(IS)=2.100, C(IS)=0.800, D(FS)=1.200

Rise/Fall: A→B = 1.500–2.100 = –0.600 (Fall); B→C = 2.100–0.800 = +1.300 (Rise); C→D = 0.800–1.200 = –0.400 (Fall)

RL: B = 50.000–0.600 = 49.400; C = 49.400+1.300 = 50.700; D = 50.700–0.400 = 50.300

Check: BS–FS = 1.500–1.200 = +0.300; Rise–Fall = 1.300–1.000 = +0.300; Last–First = 50.300–50.000 = +0.300 ✓

Advantages: (1) Complete arithmetic check including intermediate sights; (2) Errors in any reading are immediately detectable; (3) More systematic and reliable than the HI method.

2M Q10. What is magnetic declination? Give its significance.

Magnetic Declination is the horizontal angle between the geographic (true) north and the magnetic north at any given location. It varies with position on the earth and changes over time due to the movement of the magnetic pole. Significance: All compass readings give magnetic bearings; to convert them to true bearings (needed for accurate maps), the magnetic declination must be applied as a correction. True Bearing = Magnetic Bearing ± Declination (+ for westward declination, – for eastward).

Section 14: Quick Revision Sheet

⚡ **EXAM READY REVISION:** Memorize these key points for maximum marks.

🔑 Key Definitions

- **Surveying:** Science of measuring positions of points
- **Bearing:** Angle from north to a survey line (clockwise)
- **RL:** Height above datum (MSL)
- **HI:** RL of BM + BS reading
- **BM:** Fixed point of known elevation
- **Contour:** Line joining equal elevation points
- **CI:** Vertical gap between contours
- **Rise:** Prev. reading > Present reading
- **Fall:** Prev. reading < Present reading

📐 Essential Formulas

- HI = RL_BM + BS
- RL = HI – FS (or IS)
- Rise/Fall = Prev.Rdg – Pres.Rdg
- ΣBS – ΣFS = ΣRise – ΣFall = Last RL – First RL
- BB (WCB): FB ± 180°
- WCB→QB: See quadrant table
- True Bearing = Mag. Bearing ± Declination
- Slope = CI / HE
- H = L × cos θ (horizontal dist.)

🧭 WCB → QB Conversion

- 0°–90°: N (WCB) E
- 90°–180°: S (180°–WCB) E
- 180°–270°: S (WCB–180°) W
- 270°–360°: N (360°–WCB) W

🧭 Back Bearing Rule

- FB < 180°: BB = FB + 180°
- FB > 180°: BB = FB – 180°
- QB: Change N↔S, E↔W

🗺 Contour Rules (Must Know)

- Same RL on every point of a contour
- Two contours NEVER cross
- Closer = Steeper; Wider = Gentler
- Valley: V-shapes point UPSTREAM
- Ridge: U-shapes open DOWNHILL
- Hill: Closed, increasing inward
- Depression: Closed with tick marks
- Always cross ridges/valleys at 90°

📏 Levelling Methods Comparison

- **HI Method:** Fast, no check on IS
- **R&F Method:** Slow, checks ALL readings
- **Simple:** One setup, two points
- **Differential:** Multiple setups, far apart
- **Reciprocal:** Eliminates refraction/collimation errors

🧭 Instruments Quick Reference

- **Prismatic Compass:** WCB (0–360°), graduated S→N
- **Surveyor's Compass:** QB (0–90° per quadrant)
- **Dumpy Level:** Fixed telescope, robust
- **Auto Level:** Self-levelling compensator
- **Digital Level:** Bar-coded staff, electronic reading
- **Chain:** 20m/30m metric; 100 links

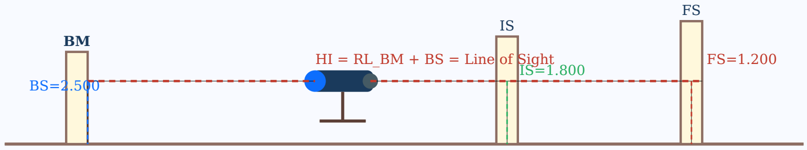
⚠ Common Errors in Levelling

- Non-vertical staff (always check plumb)
- Parallax error (improper focusing)
- Earth curvature and refraction (reciprocal levelling)
- Settlement of tripod (firm ground setup)
- Wrong identification of BS/FS/IS

✅ Checklist for Exam Problems

- ☐ Identify BS, IS, FS correctly
- ☐ Calculate HI for each instrument position
- ☐ Find RL = HI – Reading
- ☐ Apply arithmetic check at the end
- ☐ State Rise/Fall for each consecutive pair
- ☐ Label all diagram components clearly
- ☐ State the formula used before solving

QUICK REFERENCE: LEVELLING NOTATION SUMMARY



Textbook & Reference Sources

Prescribed Textbooks

T1. M.S. Palanisamy, *Basic Civil Engineering*, Tata McGraw Hill Publications. [Unit II – Surveying: Chapters 3–5]

T2. S.S. Bhavikatti, *Introduction to Civil Engineering*, New Age International Publishers, 2022. [Bearings, Levelling, Contours]

T3. Satheesh Gopi, *Basic Civil Engineering*, Pearson Education India, 2009. [Survey measurements, Chain surveying, Compass traversing]

Reference Books

R1. S.K. Duggal, *Surveying, Vol. I & II*, Tata McGraw Hill Education. [Comprehensive reference for all surveying topics]

R2. Santosh Kumar Garg, *Hydrology and Water Resources Engineering*, Khanna Publishers. [Watershed and contour mapping applications]

R3. Santosh Kumar Garg, *Irrigation Engineering and Hydraulic Structures*, Khanna Publishers. [Canal alignment surveys, level data]

R4. Khanna, Justo & Veeraraghavan, *Highway Engineering*, Nem Chand & Bros. [Profile levelling for road projects]

R5. IS 10500:2012, *Drinking Water – Specification (Second Revision)*, Bureau of Indian Standards.

Unit II – Topic-wise Study Map

Topic	Key Concept	Exam Weight	Key Formula
Introduction & Objectives	Definitions, principles	2 marks	–
Horizontal Measurements	Chain surveying, tape corrections	2–5 marks	$H = L \cos \theta$
Angular Measurements	Prismatic compass, angles	2–5 marks	Sum of angles = $(2n-4)\times 90°$
Bearings (WCB & QB)	Conversions, fore/back bearing	5–10 marks	$BB = FB \pm 180°$
Levelling Instruments	Dumpy level, staff reading	5 marks	$HI = RL + BS$
Methods of Levelling	HI method, R&F method	10 marks	$RL = HI - FS$; Check: $\Sigma BS - \Sigma FS$
Problems on Levelling	Numerical problems, tables	10 marks	All levelling formulas
Problems on Bearings	WCB↔QB, back bearing calc.	5 marks	WCB→QB quadrant rules
Contour Mapping	Properties, methods, uses	5–10 marks	Slope = CI/HE

📌 **JNTUK Exam Tips for Unit II:**
1. Always draw labeled diagrams (Dumpy Level, Contours, Bearing system) – carries 2–3 marks per diagram.
2. Numerical problems on levelling (HI and R&F method) are almost always asked for 10 marks.
3. Bearing conversions (WCB↔QB) and back bearing calculations are high-scoring short questions.
4. For contour questions, list all 9 characteristics along with a sketch showing hill vs. valley.
5. Always write the arithmetic check for levelling problems – you lose marks if omitted.

Best of Luck for Your Examinations!